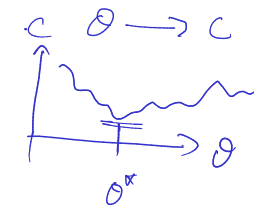


Building blocks of a variational quantum algorithm



$$\theta^* = \arg \min_{\theta} C(\theta)$$

1. Cost Function

- encodes the solution to the problem
- use classical optimizers to optimize parameters θ

$$C(\theta) = f(\{\rho_k\}, \{O_k\}, U(\theta))$$

$$C(\theta) = \sum_k f_k(\text{Tr}[O_k U(\theta) \rho_k U^\dagger(\theta)])$$

2. Ansatzes

- parameterized quantum circuit
- structure depends on problem

$$U(\theta) = U_L(\theta_L) \cdots U_2(\theta_2) U_1(\theta_1)$$

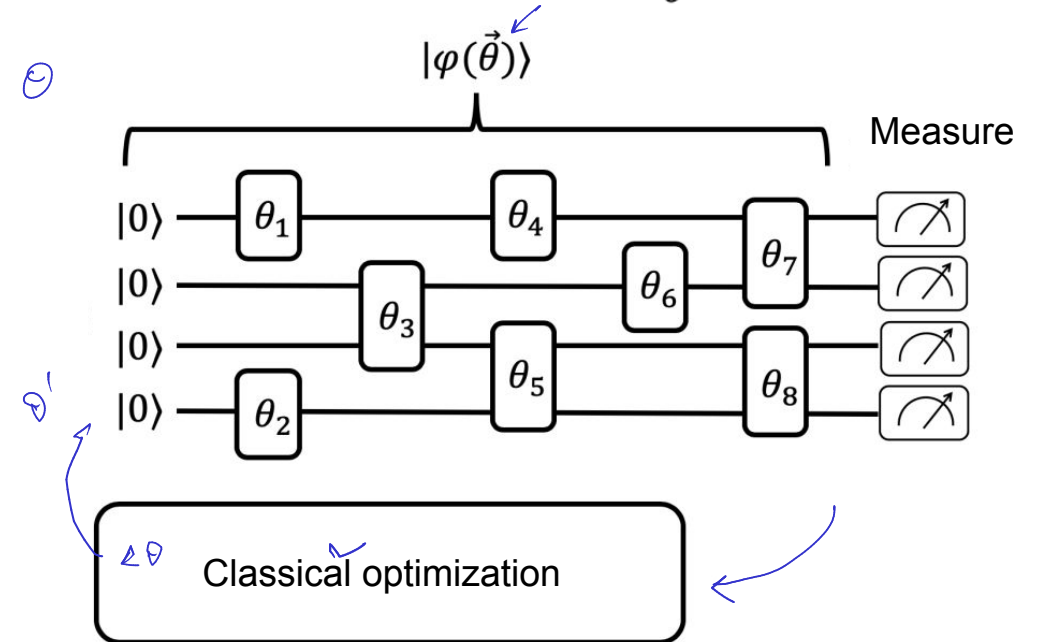
3. Gradients

- analytically evaluate the gradients $\Rightarrow$ resource-efficient

$$\frac{\partial C}{\partial \theta_l} = \sum_k \frac{1}{2 \sin \alpha} (\text{Tr}[O_k U^\dagger(\theta_+) \rho_k U(\theta_+)] - \text{Tr}[O_k U^\dagger(\theta_-) \rho_k U(\theta_-)])$$

4. Optimizers

- classical optimizers: gradient descent



"Parameter-shift trick"

$\theta \Rightarrow$ variable $\Rightarrow C(\theta)$ $\Rightarrow$ classical optimization

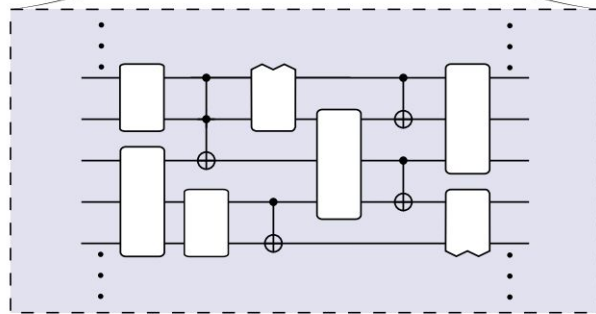
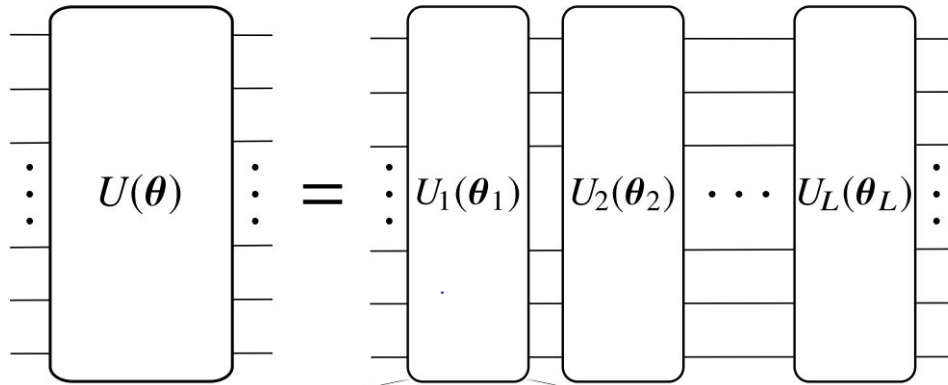
$\theta_+ = \theta + \delta\theta$ $\theta_- = \theta - \delta\theta$

Note: PS is different from finite diff method.

$(\theta + \delta\theta) \Rightarrow \delta\theta \rightarrow 0$

Building blocks of a variational quantum algorithm

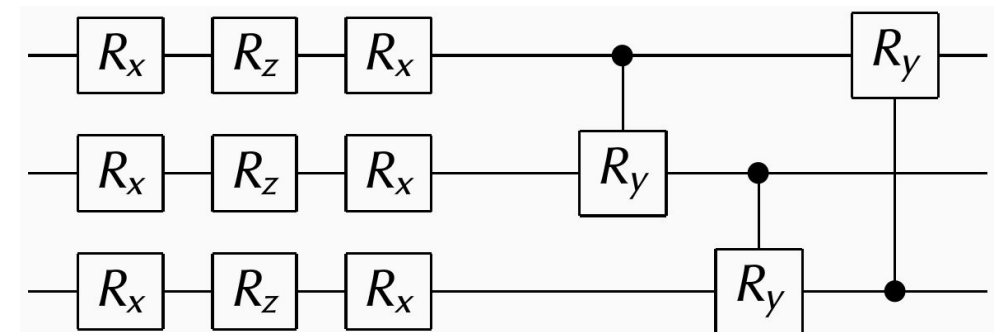
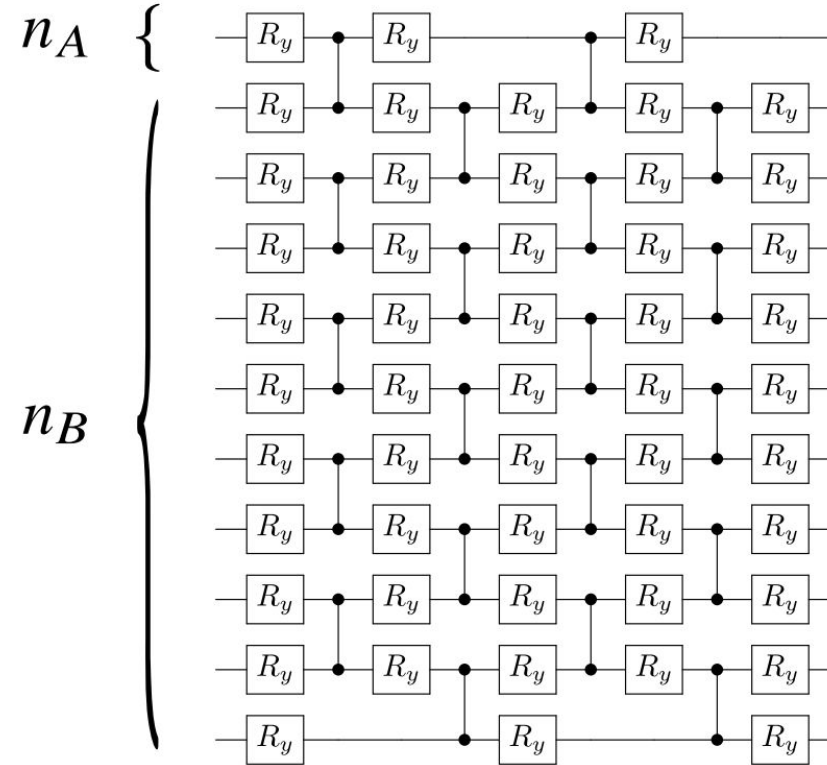
$$U(\boldsymbol{\theta}) = U_L(\boldsymbol{\theta}_L) \cdots U_2(\boldsymbol{\theta}_2) U_1(\boldsymbol{\theta}_1)$$



$$U_l(\theta_l) = \prod_m e^{-i\theta_m H_m} W_m$$

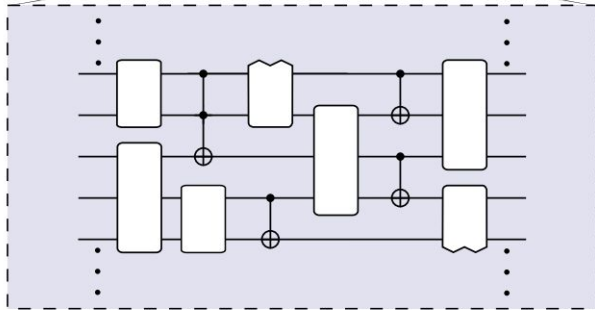
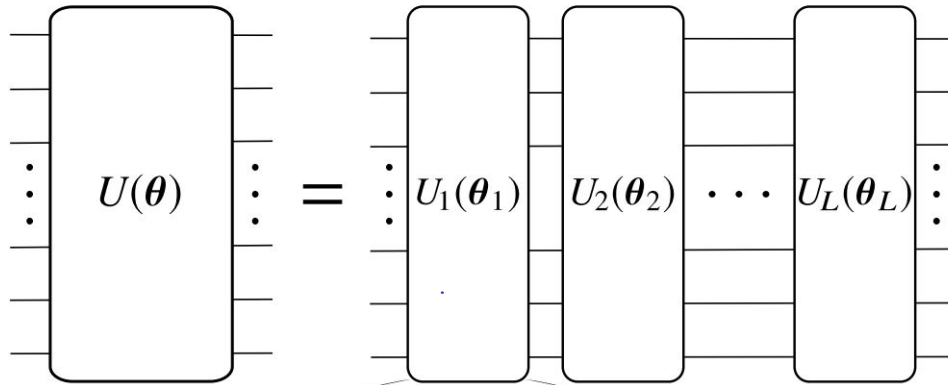
parametrization

non-parametrized
unitary



Building blocks of a variational quantum algorithm

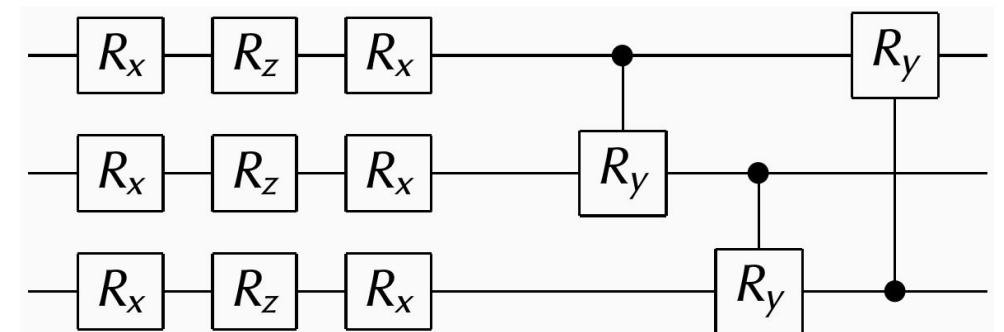
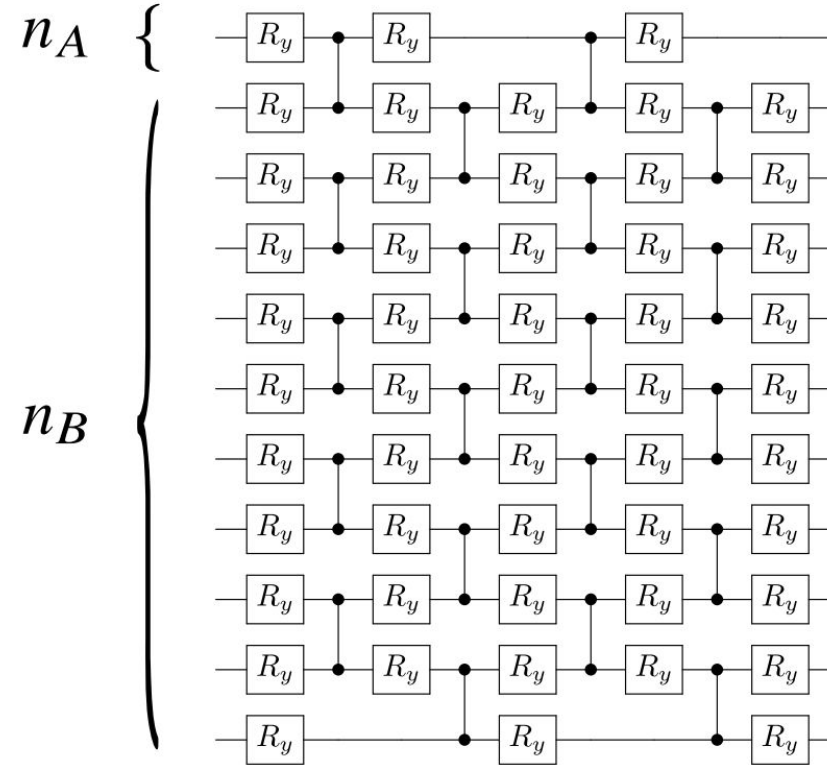
$$U(\boldsymbol{\theta}) = U_L(\boldsymbol{\theta}_L) \cdots U_2(\boldsymbol{\theta}_2)U_1(\boldsymbol{\theta}_1)$$



$$U_l(\theta_l) = \prod_m e^{-i\theta_m H_m} W_m$$

parametrization

non-parametrized
unitary



Building blocks of a variational quantum algorithm

$\boxed{A} \Rightarrow$ quantum H/W aware

NISQ $\rightarrow$ limited gate alphabet
 $\rightarrow$ limited qubit connect
 $\rightarrow$ "shallow circuits"

$n_A \Rightarrow$

1. Hardware-efficient ansatz

- aimed at reducing the circuit depth
- use gate alphabet that is hardware aware
 - connectivity and interactions

"static"
 structure is fixed

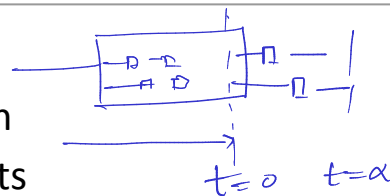
2. Problem-inspired PQC ansatz (tailored)

- Approximation to the solution by construction
- high circuit-depth or gate complexity in general
- Example: UCC ansatz $\rightarrow$ quantum chemistry applications

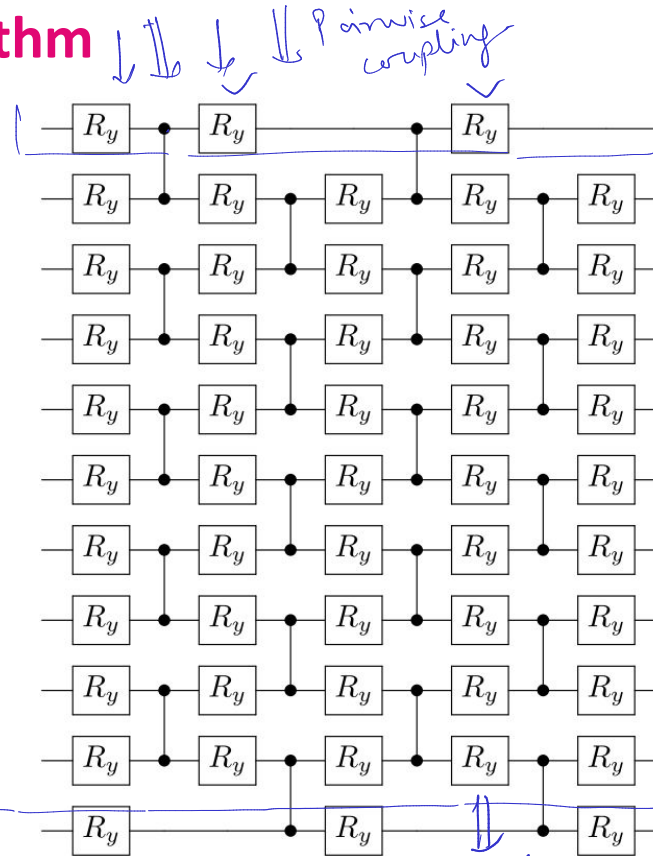
Problem specific

3. Variable circuit ansatz

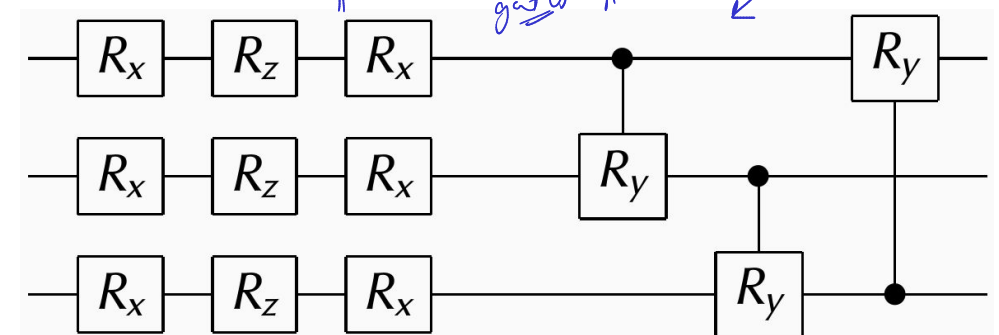
- Optimize for circuit-depth
- Construct adaptive circuits
- Example: ADAPT-VQE



n_B



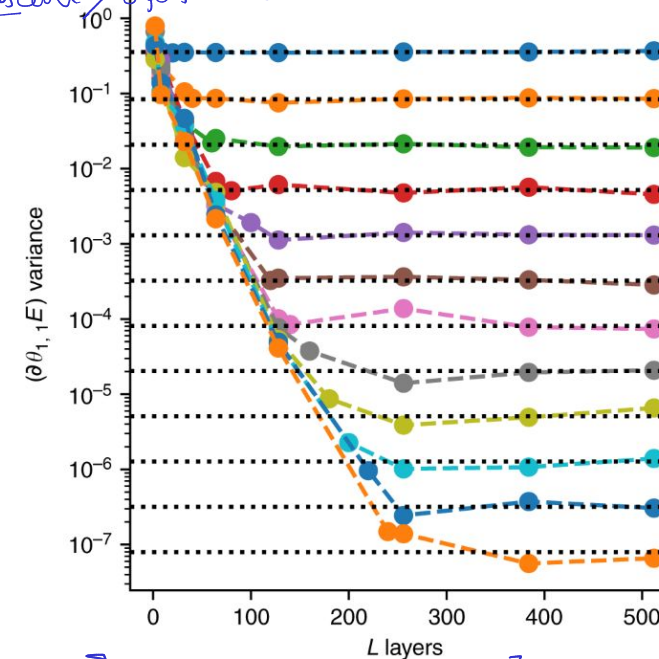
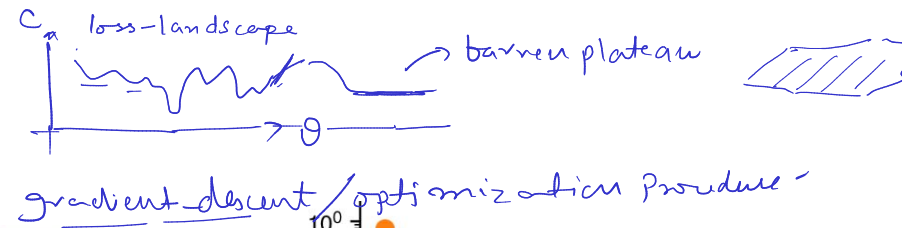
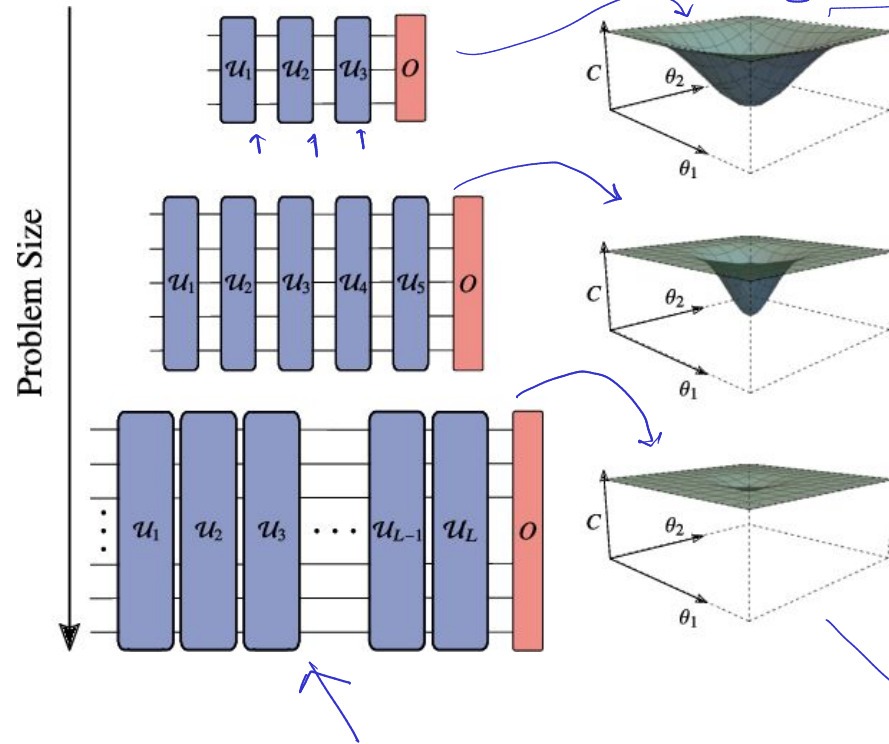
Alternating Layered Ansatz



Hardware Efficient Ansatz

Challenge (1) : Barren Plateaus

"system size" $\rightarrow$ #gates/circuit depth
 $\rightarrow$ # qubits



M. Cerezo et. al. Nature Communications 12, 1791 (2021)

1. Gradient's variance saturates at a certain depth.
2. Gradient's variance decays as number of qubits is increased.
3. Also induced by noise present in NISQ hardware.

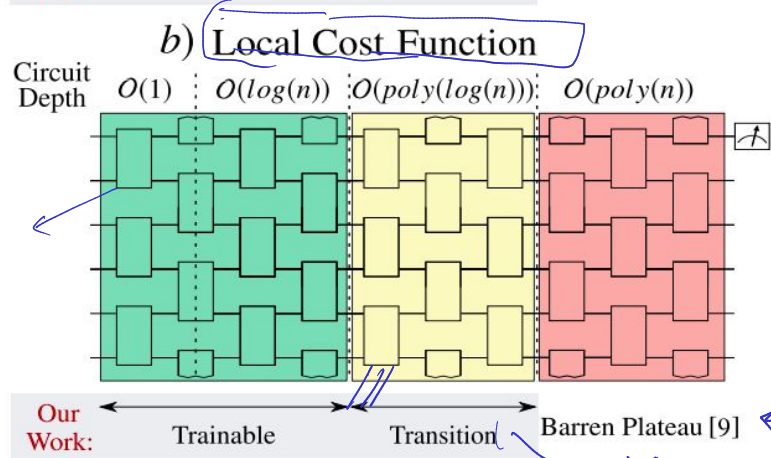
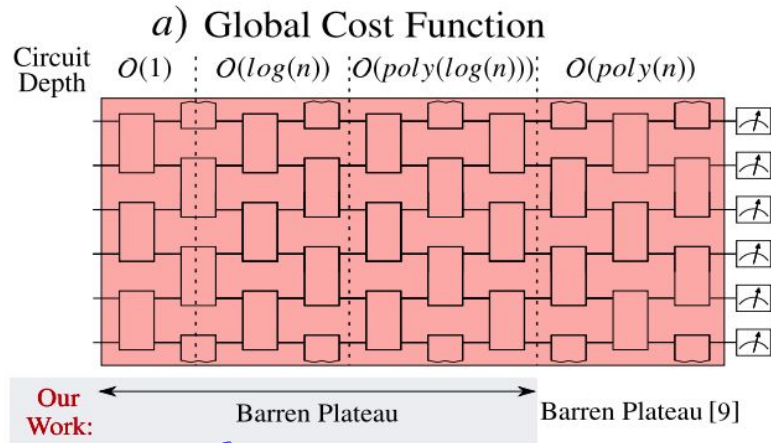
cause
"the magnitude of cost function's partial derivatives will be, on average, exponentially vanishing with the system size"

$$\langle \partial \rangle \rightarrow 0$$

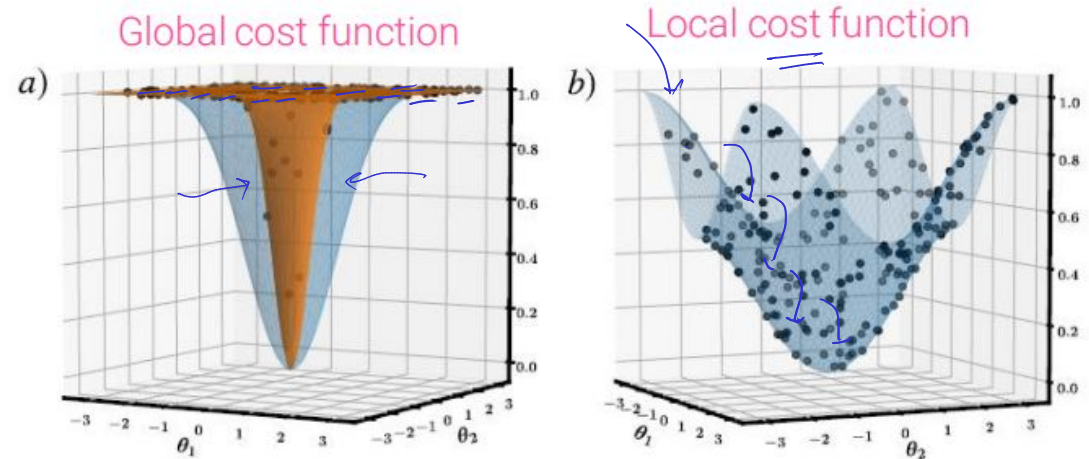
$\Delta\theta = f\left(\frac{\partial C}{\partial \theta}\right)$
 $\downarrow$
 lose variational quantum advantage
 $(\partial_{\theta_1,1} E) \downarrow$

Challenge (1) : Barren Plateaus

variational algo. have limits - Perform best



REF: Cost Function Dependent Barren Plateaus in Shallow Parametrized Quantum Circuits, M. Cerezo, et. al.

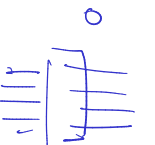


Potential Solutions to Barren Plateaus:

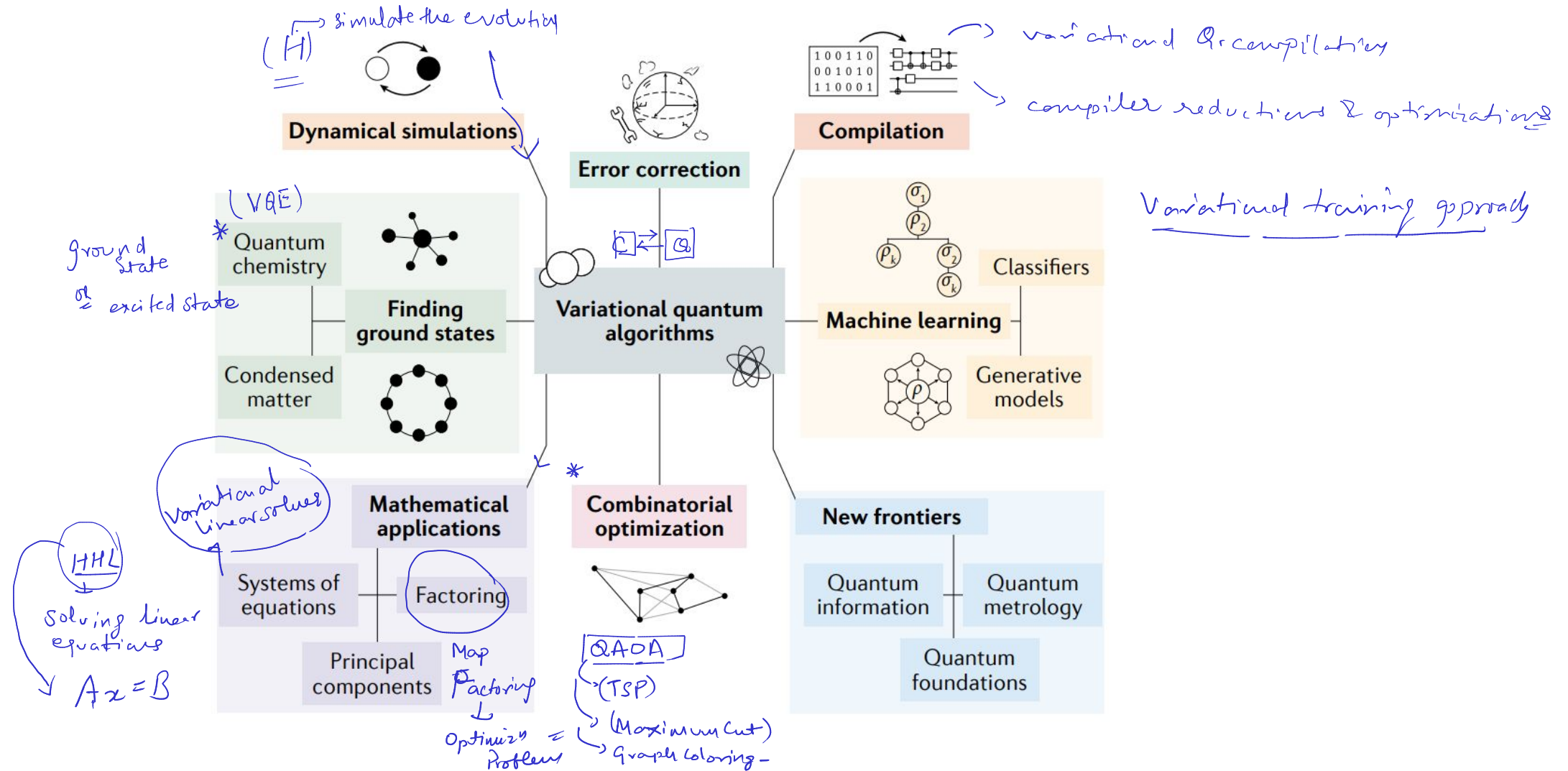
1. Use parameters close to the solution
 - Usually done by classical preprocessing
2. Use local cost functions instead of global cost functions
3. Introduce correlations between parameters

C_{min}

"hack"



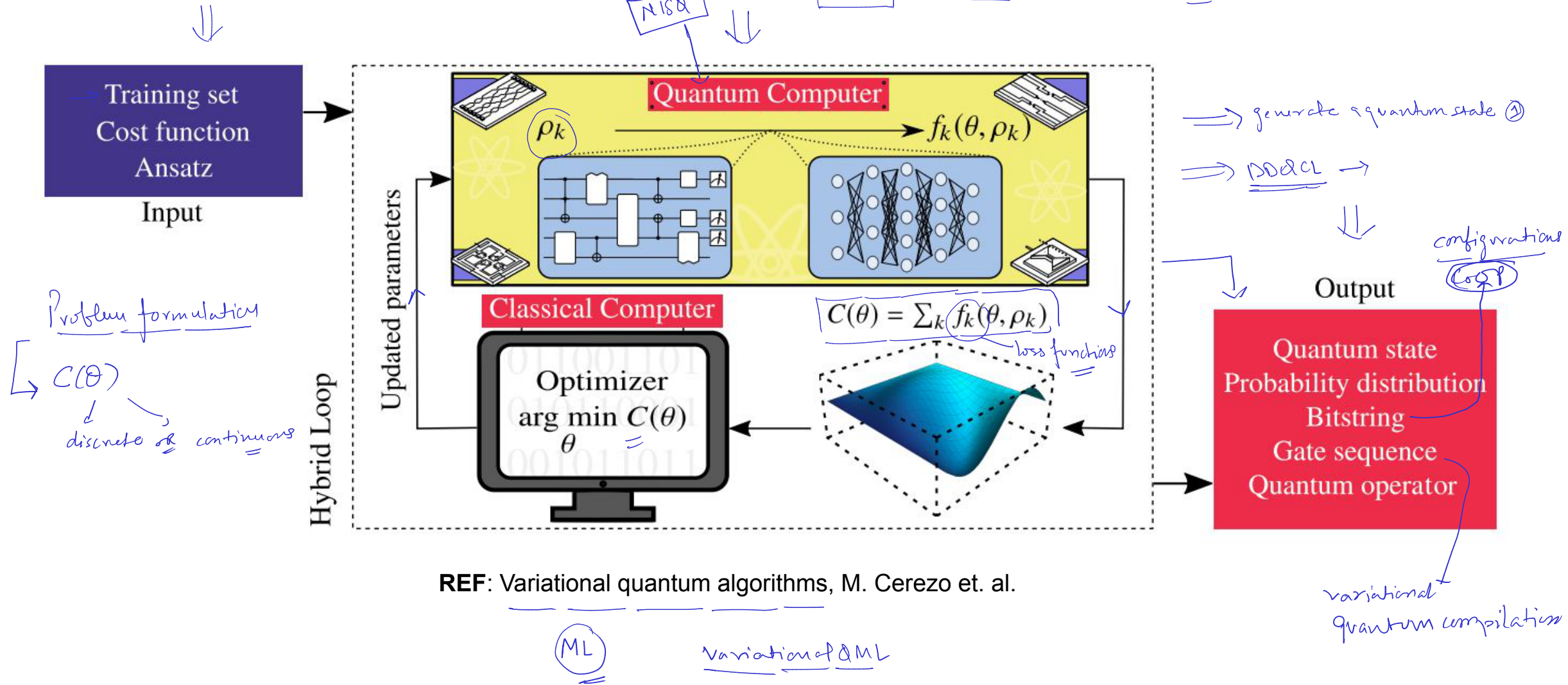
Current-generation application landscape for VQAs



REF: Variational quantum algorithms, M. Cerezo et. al.

What are we learning in this lecture?

QPU → CPU hybrid q/c scheme.



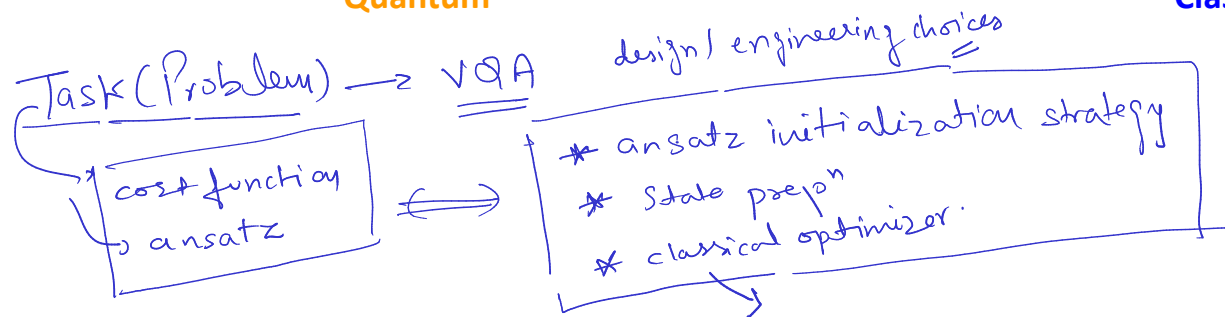
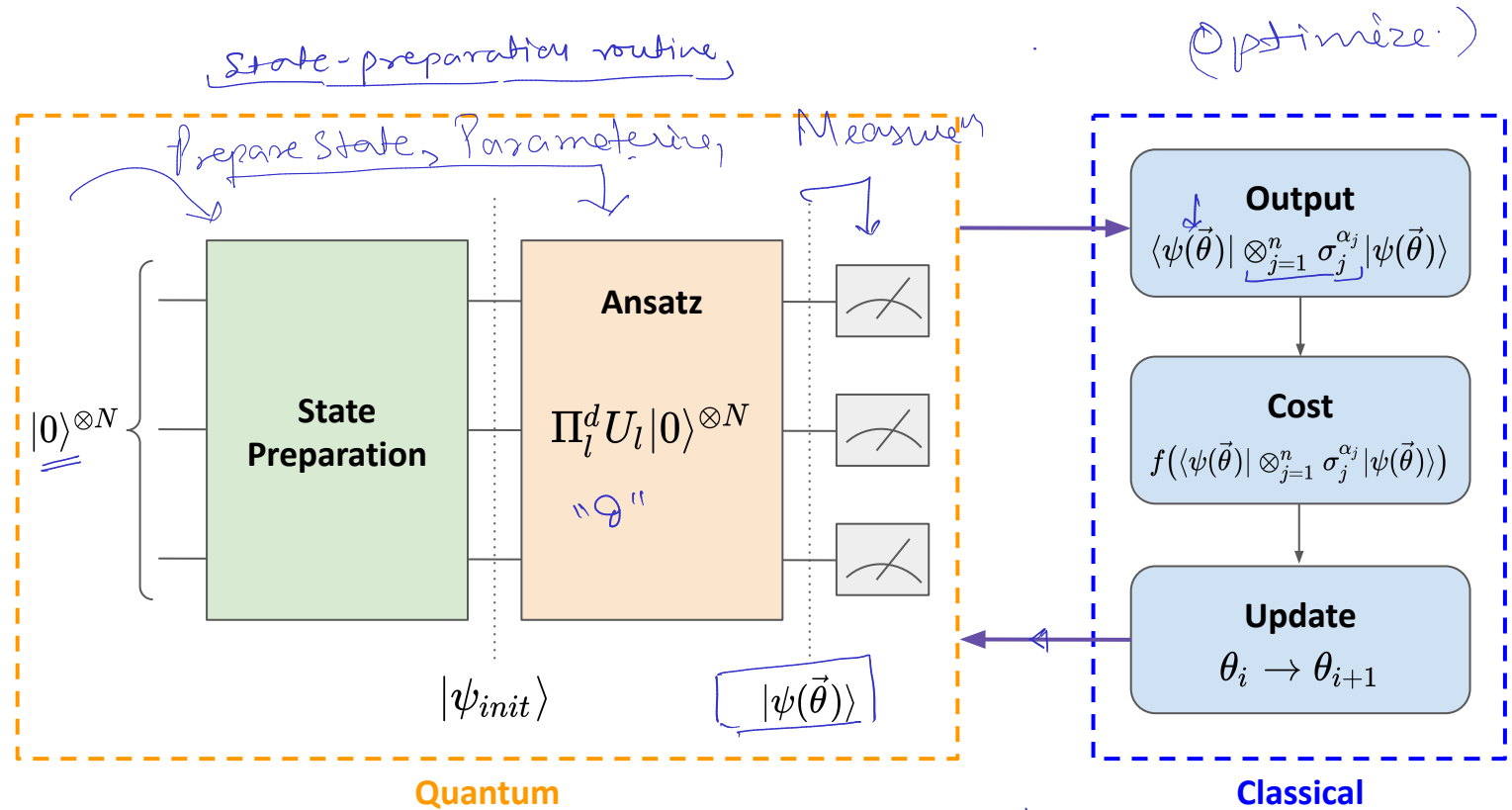
REF: Variational quantum algorithms, M. Cerezo et. al.

Setup: Variational quantum algorithm (VQA)

Task: Minimize a cost function defined by a Hamiltonian $(\hat{H}), C(\theta)$

Elements:

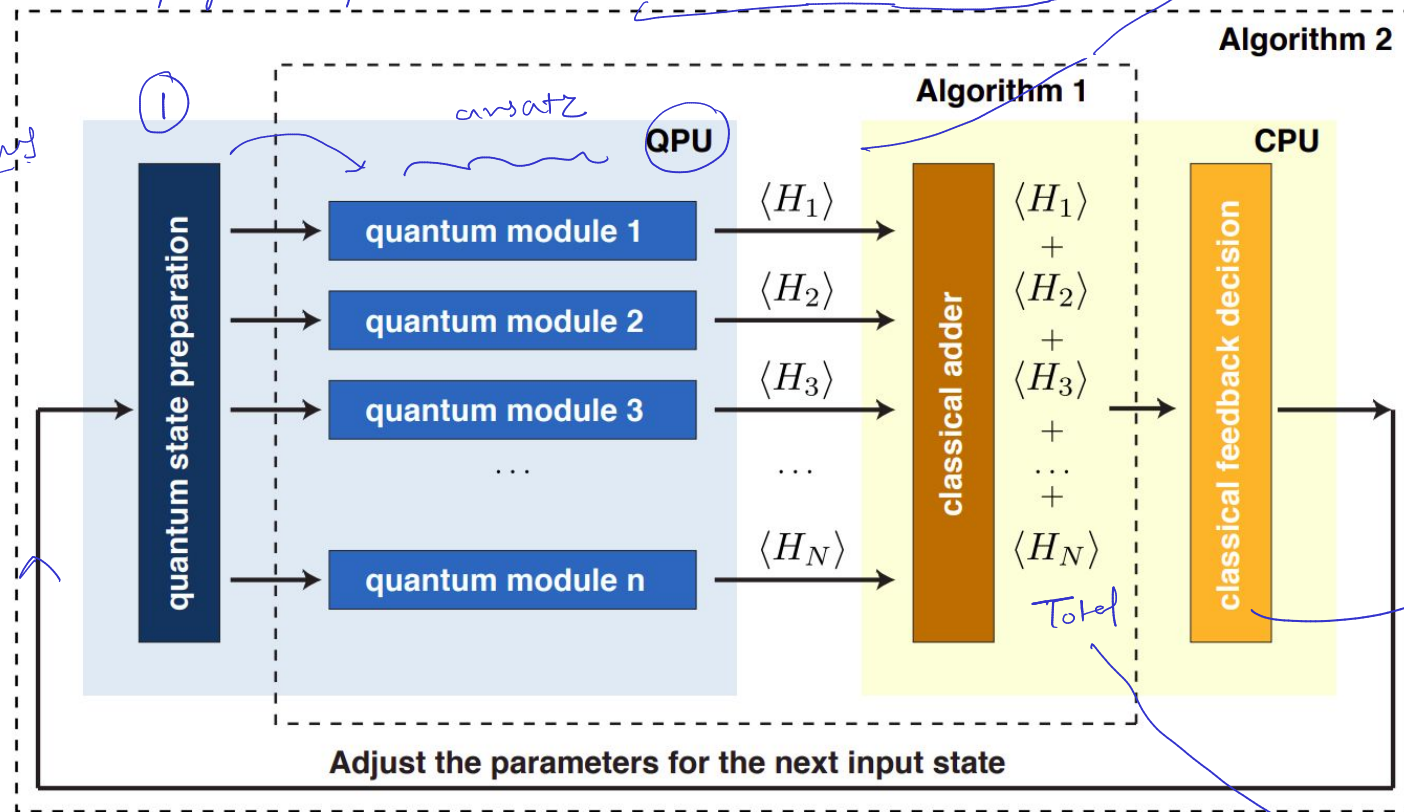
1. Variational Parameters θ
2. Ansatz d -depth PQC
3. Parameterized State
4. Measurements
5. Cost Function
6. Optimization Loop



VQA Example (1): Variational Quantum Eigensolver

Example: ground-state predⁿ
 chemical / Molecular Systems
 $[a] \rightleftharpoons [c] \rightarrow CO$
 ADAPT VQE
 ansatz

Molecular (H) $\xrightarrow{\text{computational representation}}$ $H = \sum h_{\alpha} \bigotimes_{j=1}^N (\sigma_j^{\alpha_j})$ ✓
 physical representation
 qubit hamiltonian



Parameterized Ansatz

$$\frac{\langle \psi(\vec{\theta}) | H | \psi(\vec{\theta}) \rangle}{\langle \psi(\vec{\theta}) | \psi(\vec{\theta}) \rangle} \approx E_0$$

updated $\Rightarrow$

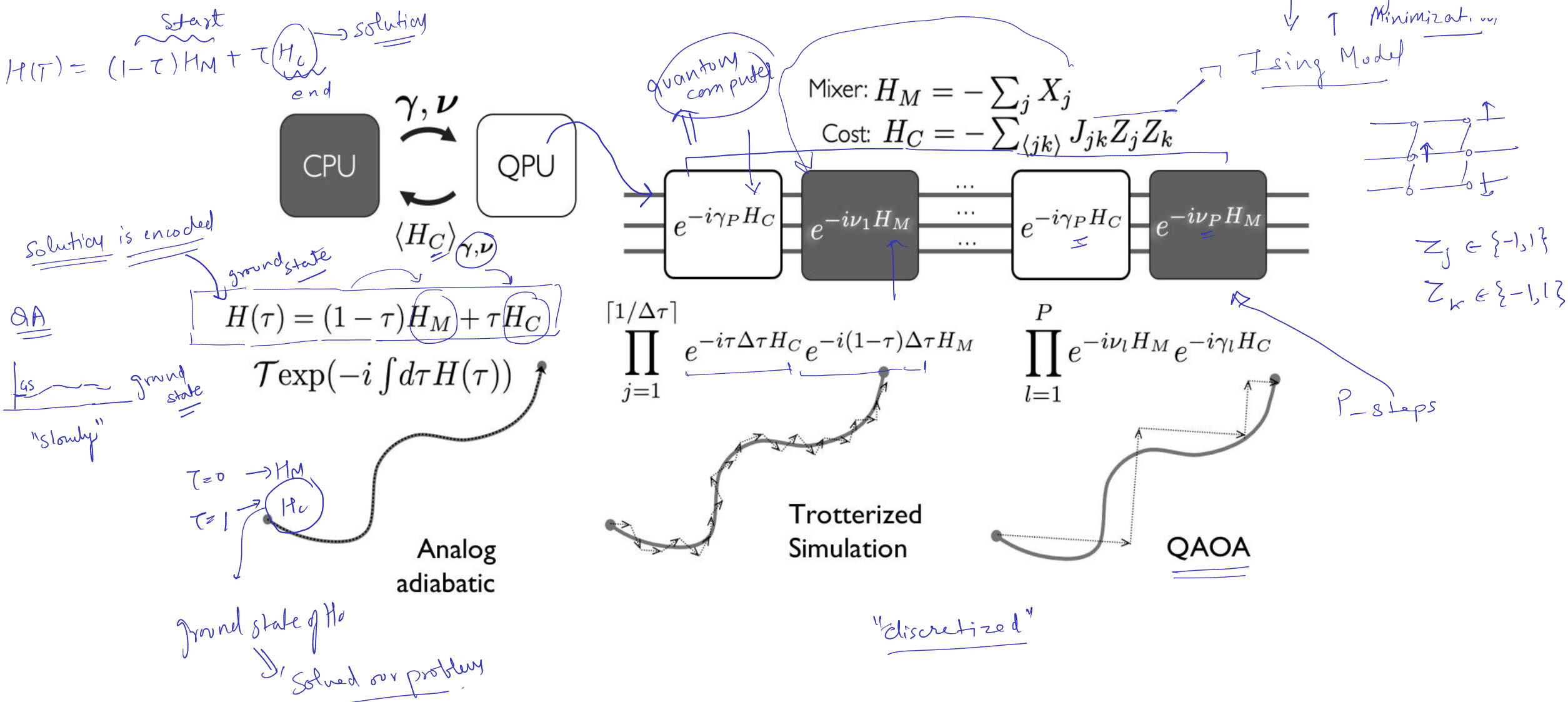
$$\theta_i \rightarrow \theta_{i+1}$$

$$\langle \psi(\vec{\theta}) | \bigotimes_{j=1}^N \sigma_j^{\alpha_j} | \psi(\vec{\theta}) \rangle$$

Energy computation

REF: A variational eigenvalue solver on a quantum processor, Alberto Peruzzo, et. al.

$Q\pi_{AI}$ VQA Example (1): Quantum Approximate Optimization Algorithm



Summary

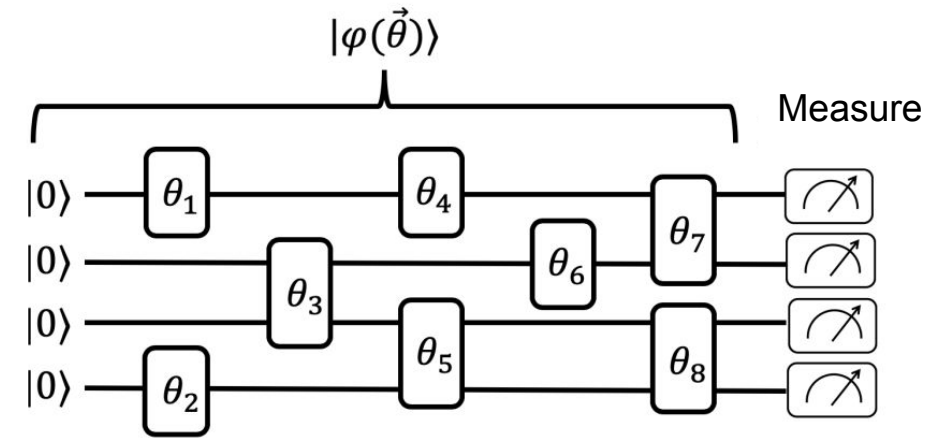
Variational Quantum Algorithms

1. Uses hybrid quantum-classical model
2. Optimizes short depth ^{shallow} circuits for optimal use
3. Agnostic to systematic errors
4. Hardware-aware implementations
5. Hardware-software co-design ^{→ better slw models} _{→ better devices}

challenges

- Trainability
 - Barren Plateaus
 - Parameter Initialization ^{→ random} _{bad}
 - Ansatz Structure ^{hlw} _{"Ansatz Search"}
 - Efficiency
 - Resource utilization
 - Circuit compilations
 - Accuracy ^{Noise}
 - NISQ Noise Modeling
 - Quantum Error Mitigation ✓
- $\downarrow$
NAS RL
—
GV

$$\theta^* = \arg \min_{\theta} C(\theta)$$



Classical optimization